

UNIT TEST REPORT

By Firmgen

Report identity

Project	vehicle_monitor
Target	firmware (verified off target, no MCU hardware in the loop)
Report ID	UTR-E97CA512
Run ID	e97ca512-53fa-4831-9782-25279f994e3d
Execution date	24 September 2026, 09:14 UTC
Prepared by	Firmgen automated unit-test agent
Cases executed	129 across 39 test file(s)

1. Test Objective

This activity established 81 passing cases out of 129, with executed evidence for 1 of 19 traced requirements and verification of 20 of 58 enumerated firmware units. It exercised 39 test files across climate sensing, platform and power, threshold evaluation, WiFi and MQTT communication, diagnostics and recovery, and MPU6050 motion sensing and I2C driver code, supporting focused follow-up on the remaining cases and units.

2. Environment and Scope

Source under test	vehicle_monitor
Compile-time defines	UNIT_TEST, HOST_TEST, UNITY_INCLUDE_EXEC_TIME
Test parallelism	8 thread(s)
Test framework	Ceedling 1.1.6 (Unity + CMock)
Build system	Ceedling 1.1.6
Compiler	C:\Users\YatendraBhargava\AppData\Local\craftifai\toolchains\mingw\16.2.0posix-14.0.0-ucrt-r1\bin\gcc.exe 16.2.0
Host environment	FirmGen-managed toolchain (ruby 3.3.12-1) on Windows-10
Execution mode	Off target - compiled for the build architecture, no MCU hardware in the loop; executed directly on the build host, not in a container
Test cases executed	129 across 39 test file(s)

3. Test Harness and Reproducibility

The tests were built and run with Ceedling 1.1.6 using Unity and CMock, with UNIT_TEST, HOST_TEST, and UNITY_INCLUDE_EXEC_TIME defined. They ran in parallel on 8 threads using GCC 16.2.0 in the FirmGen-managed Ruby 3.3.12-1 Windows-10 host environment. Execution was off target on the build architecture, directly on the host, with no MCU hardware in the loop; CMock provided the mock-based test support used by the harness.

Reproduction step	Command	Result
Configure	`ceedling project.yml (generated per run)`	● PASS
Build and link	`ceedling test:build_only`	● PASS
Execute tests	`ceedling test:all`	● PASS

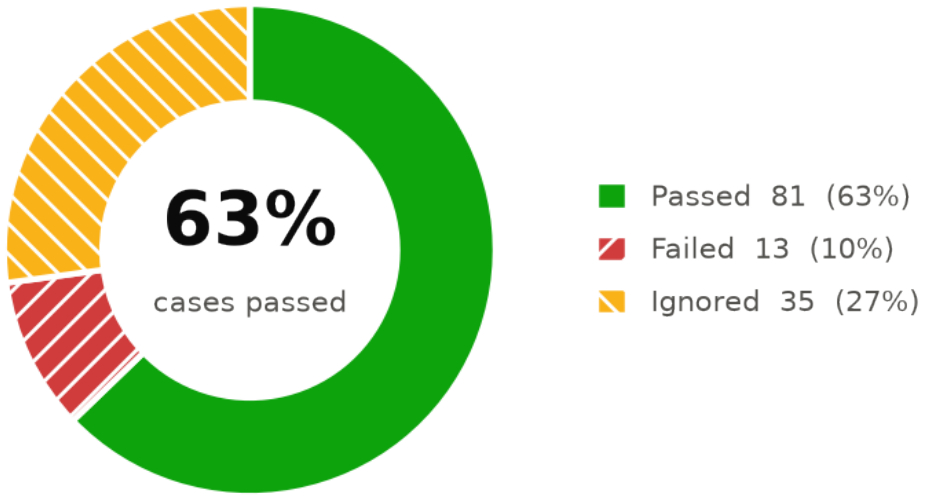
Evidence shipped with this run: executed_tests.xlsx, run.txt

4. Execution Summary

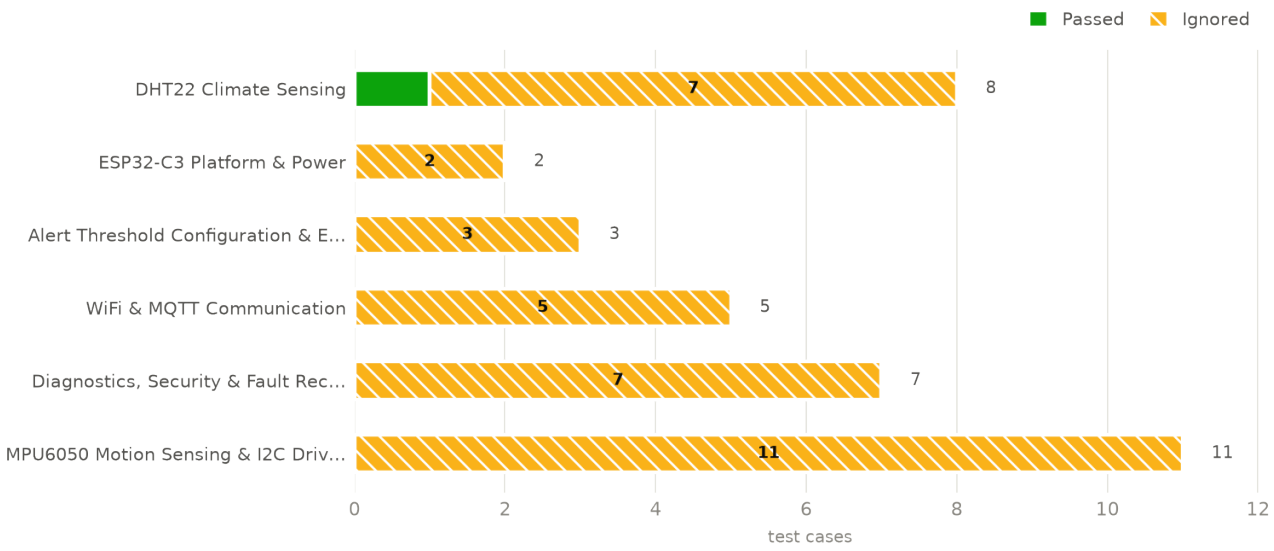
81	63%	129	13
Cases passed	Case pass rate	Cases executed	To address

Metric	Result
Cases passed	81
Case pass rate	62.8%
Requirements fully passing	1
Requirements traced to test evidence	19
Total test cases	129
Test files executed	39
Requirement-driven cases	36
Firmware-coverage cases	93
Cases awaiting a verdict	35
Cases with a failure to address	13
Requirements awaiting a verdict	18
Requirements with a failure to address	0
Total execution time	23,188 ms

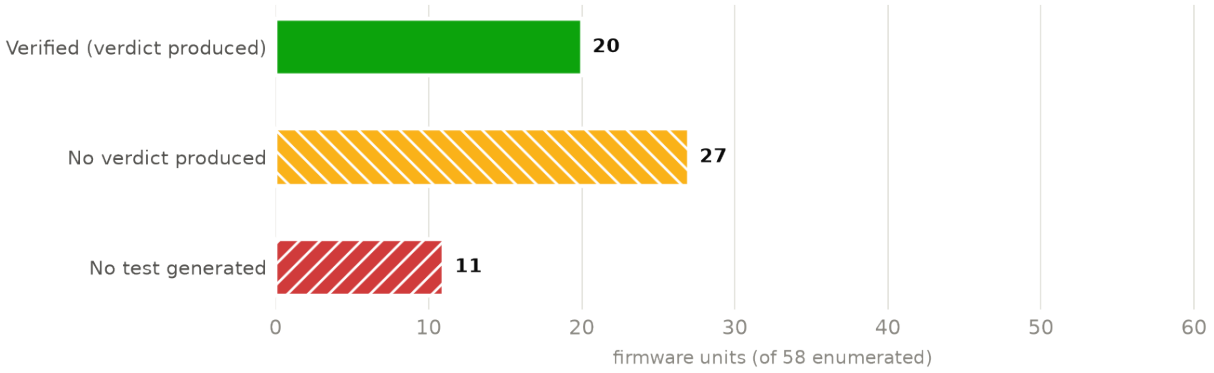
Outcome of every executed case



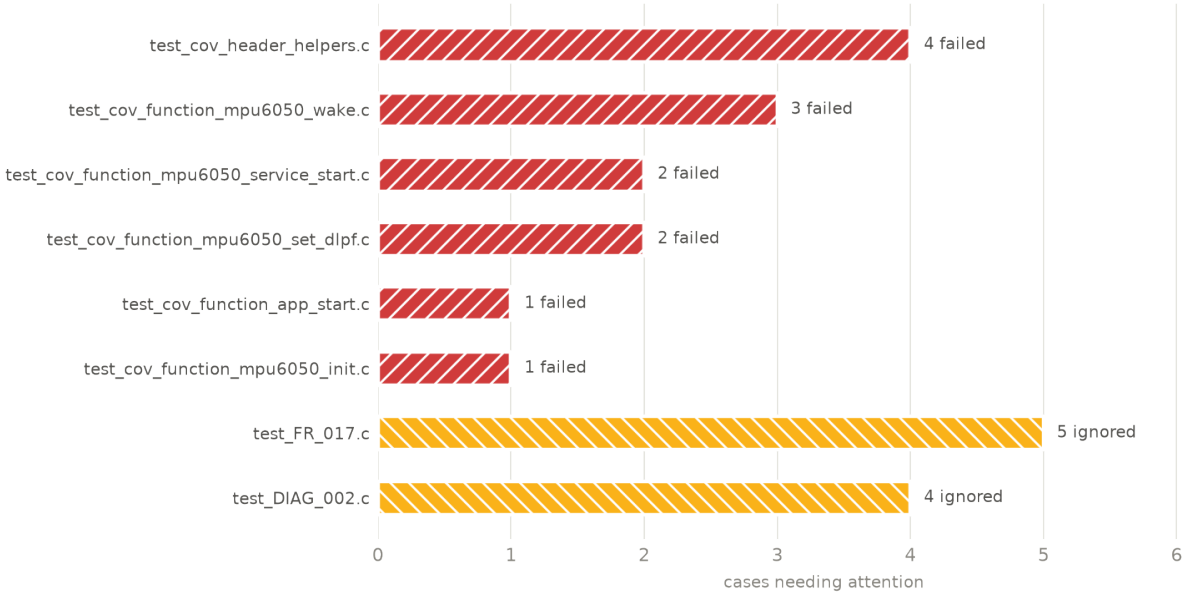
Outcomes by requirement category, strongest first



Firmware units verified by this suite



Where to focus next



5. Traceability

The executed evidence covers 36 requirement cases and records 1 of 19 requirements as passing; 18 requirements remain not yet verified, with 0 requirements classified as requiring an assertion fix. The remaining untraced portion reflects cases for which the supplied implementation, interfaces, acceptance criteria, or hardware evidence did not provide a decidable result.

Requirements traced to executed cases

Req. ID	Category	Module / symbol	Test file	Cases	Result
FR-003	DHT22 Climate Sensing	dht22_checksum_valid +2 more	test_FR_003.c	1	PASSED
DIAG-002	Diagnostics, Security & Fault Rec…	dht22_service_sample +3 more	test_DIAG_002.c	4	IGNORED
FR-004	MPU6050 Motion Sensing & I2C Driv…	mpu6050_read_sample +1 more	test_FR_004.c	2	IGNORED
FR-005	MPU6050 Motion Sensing & I2C Driv…	mpu6050_read_sample +1 more	test_FR_005.c	1	IGNORED
FR-006	MPU6050 Motion Sensing & I2C Driv…	mpu6050_read_sample +1 more	test_FR_006.c	1	IGNORED
FR-015	WiFi & MQTT Communication	mqtt_event_handler +2 more	test_FR_015.c	2	IGNORED
FR-016	DHT22 Climate Sensing	dht22_checksum_valid +2 more	test_FR_016.c	4	IGNORED
FR-017	MPU6050 Motion Sensing & I2C Driv…	mpu6050_initialize +1 more	test_FR_017.c	5	IGNORED
HW-001	ESP32-C3 Platform & Power	app_start	test_HW_001.c	1	IGNORED
HW-002	DHT22 Climate Sensing	dht22_port_open +1 more	test_HW_002.c	1	IGNORED
HW-004	ESP32-C3 Platform & Power	app_start	test_HW_004.c	1	IGNORED
IF-002	WiFi & MQTT Communication	network_event_handler +1 more	test_IF_002.c	1	IGNORED
IF-007	DHT22 Climate Sensing	dht22_port_open +1 more	test_IF_007.c	1	IGNORED
IF-008	DHT22 Climate Sensing	dht22_port_open +1 more	test_IF_008.c	1	IGNORED
IF-011	MPU6050 Motion Sensing & I2C Driv…	mpu6050_initialize +1 more	test_IF_011.c	1	IGNORED
IF-012	MPU6050 Motion Sensing & I2C Driv…	mpu6050_initialize +1 more	test_IF_012.c	1	IGNORED
IF-013	WiFi & MQTT Communication	mqtt_event_handler +2 more	test_IF_013.c	2	IGNORED
NFR-002	Alert Threshold Configuration & E…	threshold_service_load +1 more	test_NFR_002.c	3	IGNORED
SF-001	Diagnostics, Security & Fault Rec…	app_start, diagnostics_task	test_SF_001.c	3	IGNORED

Firmware units traced to executed cases

Firmware unit	Kind	Test file	Cases	Result
app	module	test_cov_module_app.c	2	PASSED
board_name	function	test_cov_function_board_name.c	1	PASSED
dht22_service	module	test_cov_module_dht22_service.c	7	PASSED
dht22_service_sample	function	test_cov_function_dht22_service_sample.c	1	PASSED
dht22_service_start	function	test_cov_function_dht22_service_start.c	4	PASSED
mpu6050	module	test_cov_module_mpu6050.c	6	PASSED
mpu6050_configure_interrupt_pin	function	test_cov_function_mpu6050_configure_interrupt_pin.c	8	PASSED
mpu6050_deinit	function	test_cov_function_mpu6050_deinit.c	2	PASSED
mpu6050_get_who_am_i	function	test_cov_function_mpu6050_get_who_am_i.c	1	PASSED
mpu6050_handle_size	function	test_cov_function_mpu6050_handle_size.c	1	PASSED
mpu6050_read_fifo_count	function	test_cov_function_mpu6050_read_fifo_count.c	1	PASSED
mpu6050_read_register	function	test_cov_function_mpu6050_read_register.c	1	PASSED
mpu6050_service	module	test_cov_module_mpu6050_service.c	15	PASSED
mpu6050_service_state	function	test_cov_function_mpu6050_service_state.c	1	PASSED
app_start	function	test_cov_function_app_start.c	1	FAILED
helpers	header	test_cov_header_helpers.c	15	FAILED
mpu6050_init	function	test_cov_function_mpu6050_init.c	16	FAILED
mpu6050_service_start	function	test_cov_function_mpu6050_service_start.c	5	FAILED
mpu6050_set_dlpf	function	test_cov_function_mpu6050_set_dlpf.c	2	FAILED
mpu6050_wake	function	test_cov_function_mpu6050_wake.c	3	FAILED

6. Failure, Ignored Case, and Root-Cause Record

Scope / cases	Observed result	Suspected cause	Severity	Disposition
Hardware identity, power, and wiring evidence HW-001, HW-002, HW-004, IF-007, IF-008, IF-011, IF-012	These cases were ignored because the archive contained no BOM, physical-board, schematic, wiring, power measurement, or firmware-visible equivalent evidence for the requested hardware criteria.	These criteria require hardware or design artifacts that are outside the supplied host-test inputs.	● Low	Open; provide the applicable BOM, board markings, schematic or wiring records, and power measurements, then perform the corresponding evidence review.
Application startup assertion app_start	test_app_start_01 recorded 2 where the test expected 3.	The startup test expectation and observed call or state count do not currently agree; the startup sequence and test setup need to be compared.	● Medium	Open; inspect the expected startup sequence and the host test setup, then rerun test_app_start_01 with the intended assertion.
Helper header assertions helpers	test_helpers_04 recorded -3 instead of 2, test_helpers_08 recorded 2 instead of 1, test_helpers_10 recorded 2 instead of -3, and test_helpers_15 recorded 6 instead of 5.	The helper tests contain multiple expected-value mismatches. The helper calculations, input fixtures, and expected values should be reviewed together before changing either code or tests.	● Medium	Open; reconcile the helper definitions and fixtures with the four expected results, then rerun the helper test cases.
MPU6050 initialization assertion mpu6050_init	test_mpu6050_init_04 recorded 0 where the test expected 1.	The initialization path did not produce the expected result under the current host test setup; the identity, configuration, and return-state assumptions need to be checked.	● Medium	Open; inspect the initialization fixture and expected return state, then rerun test_mpu6050_init_04.
MPU6050 service-start interactions mpu6050_service_start	test_mpu6050_service_start_03 called mpu6050_port_write more times than expected, while test_mpu6050_service_start_05 observed a 100 ms delay where the test expected 1000 ms.	The service-start interaction and timing expectations do not match the observed host execution. The intended register-write sequence and delay value need to be confirmed against the implementation.	● Medium	Open; align the service-start fixture and assertions with the intended write and delay sequence, then rerun both cases.
Diagnostics counter and serial-query observability DIAG-002	All four DIAG-002 cases were ignored because the available archive exposed counter accessors and a logging call, but no reachable counter-update paths or host-observable serial or log capture for the acceptance criteria.	The required counter-update and serial-query behavior is not observable through the supplied host-test interfaces.	● Medium	Open; expose controllable update paths and host-observable serial or log capture, then execute the four DIAG-002 cases.

Scope / cases	Observed result	Suspected cause	Severity	Disposition
MPU6050 published-state observability and acceptance criter… FR-004, FR-005, FR-006, FR-017	FR-004 could not observe the file-scope acceleration state, while FR-005, FR-006, and FR-017_05 lacked numeric tolerances or thresholds needed to judge conversion and axis-isolation behavior. Other FR-017 cases were also not decidable from the supplied implementation and observable interfaces.	The host tests lack access to the published MPU6050 state and the acceptance criteria do not consistently define measurable limits for the required outputs.	● Medium	Open; add an accessor or recording seam for the MPU6050 state, define numeric tolerances and axis limits, and provide observable identity, configuration readback, error, and retry…
DHT22 retry and failure-state observability FR-016	All four FR-016 cases were ignored because the private sample state and error-event mechanism were not exposed through an accessor or observable call.	The retry, retained-value, error-event, and failure-counter outcomes cannot be decided with the supplied host-test interfaces.	● Medium	Open; expose the sample outcome and error-event or counter seam needed by the acceptance criteria, then execute the four FR-016 cases.
MQTT reconnect and endpoint behavior observability FR-015, IF-002, IF-013	The FR-015 cases lacked observable reconnect, scheduling, sensor-processing, and backoff behavior. IF-002 could not inspect retained WiFi credentials, and IF-013 lacked controllable endpoint storage and observable retry or processing behavior.	The supplied host fakes and implementation excerpts do not expose enough state or behavior to decide the communication requirements.	● Medium	Open; add retained configuration and event or retry observability, including credential and endpoint inspection, then execute the communication cases.
Threshold persistence behavior observability NFR-002	All three NFR-002 cases were ignored because the archive did not provide host-side fakes or mockable interfaces for NVS operations, interruption and reboot behavior, or ESP_LOGE.	The persistence and recovery outcomes are not observable through the supplied test interfaces.	● Medium	Open; add controllable NVS, interruption, reboot, and error-log seams, then execute the three persistence cases.
Reset and recovery behavior observability SF-001	All three SF-001 cases were ignored because the supplied implementation had no observable path for automatic reboot, service resumption, reset-rate limiting, or persistent fault recording.	The host-test surface does not expose the reset and recovery behavior required by the acceptance criteria.	● Medium	Open; provide a controllable reset or watchdog recovery seam and observable service-resumption and fault-recording outcomes, then execute the three SF-001 cases.

Scope / cases	Observed result	Suspected cause	Severity	Disposition
MPU6050 DLPF host-test crashes mpu6050_set_dlpf	Both test_mpu6050_set_dlpf_01 and test_mpu6050_set_dlpf_02 crashed during execution.	The current host test setup reaches a crash before producing a test verdict. The failing setup, fake interaction, and DLPF path require isolation.	High	Open; reproduce each case with focused setup diagnostics, correct the host-test seam or fixture, and rerun both DLPF cases.
MPU6050 wake host-test crashes mpu6050_wake	test_mpu6050_wake_01, test_mpu6050_wake_02, and test_mpu6050_wake_03 crashed during execution.	The current host test setup reaches a crash in the wake path before producing a verdict. The wake-path dependencies and fake interactions require isolation.	High	Open; reproduce the wake cases with focused setup diagnostics, repair the test seam or fixture, and rerun all three cases.

7. Limitations and Exclusions

This report states what the executed cases established. The boundaries below mark where off-target unit testing ends and other verification activities begin - they are the scope of this activity, not defects in it.

- This off-target host activity does not establish electrical behavior, board identity, sensor wiring, power measurements, or physical motion response.
- It does not establish interrupt latency or RTOS scheduling behavior on the target platform.
- It does not establish MCU-specific watchdog, brown-out, reboot, WiFi, MQTT, or hardware-driver behavior without corresponding target or integration evidence.
- The activity does not provide a release decision; it records the executed unit-test evidence and the follow-up needed to obtain remaining verdicts.

8. Verification Status and Conclusion

Decision field	Conclusion
Verified scope	81 of 129 executed case(s) passed, covering 1 requirement(s) in full across 39 test file(s).
Outstanding items	Add the missing host-test observability and acceptance tolerances for diagnostics, MPU6050, DHT22, communication, persistence, and reset behavior; provide the required hardware and wiring evidence; and address the 13 cases with assertion mismatches or crashes before rerunning the 35 cases that are…
Conclusion	The strongest established result is 81 passing cases, including 1 passing case in DHT22 Climate Sensing, with 20 firmware units verified and 39 test files executed. The evidence is not yet complete across the traced requirements: 1 of 19 requirements is passing, 18 are not yet verified, and 13 cases require assertion or host-test follow-up. The next work is well defined by the observed mismatches, crashes, missing observability, undefined tolerances, and absent hardware artifacts.

Generated automatically from the executed test results of run e97ca512-53fa-4831-9782-25279f994e3d. The figures in this document are computed from executed_tests.xlsx; no count in it was written by a language model.

9. Appendix: Complete Case Record

Every case this run executed appears below, one row each - 129 in total, R1-R36 from the requirement suite and C1-C93 from the firmware-coverage suite. These are the rows every figure in Section 4 was counted from.

Case	Identifier	Test function	Result
R1	DIAG-002	test_DIAG_002_01	IGNORED
R2	DIAG-002	test_DIAG_002_02	IGNORED
R3	DIAG-002	test_DIAG_002_03	IGNORED
R4	DIAG-002	test_DIAG_002_04	IGNORED
R5	FR-003	test_FR_003_01	PASSED
R6	FR-004	test_FR_004_01	IGNORED
R7	FR-004	test_FR_004_02	IGNORED
R8	FR-005	test_FR_005_01	IGNORED
R9	FR-006	test_FR_006_01	IGNORED
R10	FR-015	test_FR_015_01	IGNORED
R11	FR-015	test_FR_015_02	IGNORED
R12	FR-016	test_FR_016_01	IGNORED
R13	FR-016	test_FR_016_02	IGNORED
R14	FR-016	test_FR_016_03	IGNORED
R15	FR-016	test_FR_016_04	IGNORED
R16	FR-017	test_FR_017_01	IGNORED
R17	FR-017	test_FR_017_02	IGNORED
R18	FR-017	test_FR_017_03	IGNORED
R19	FR-017	test_FR_017_04	IGNORED
R20	FR-017	test_FR_017_05	IGNORED
R21	HW-001	test_HW_001_01	IGNORED
R22	HW-002	test_HW_002_01	IGNORED
R23	HW-004	test_HW_004_01	IGNORED
R24	IF-002	test_IF_002_01	IGNORED
R25	IF-007	test_IF_007_01	IGNORED
R26	IF-008	test_IF_008_01	IGNORED
R27	IF-011	test_IF_011_01	IGNORED
R28	IF-012	test_IF_012_01	IGNORED
R29	IF-013	test_IF_013_01	IGNORED
R30	IF-013	test_IF_013_02	IGNORED
R31	NFR-002	test_NFR_002_01	IGNORED
R32	NFR-002	test_NFR_002_02	IGNORED
R33	NFR-002	test_NFR_002_03	IGNORED
R34	SF-001	test_SF_001_01	IGNORED
R35	SF-001	test_SF_001_02	IGNORED

Case	Identifier	Test function	Result
R36	SF-001	test_SF_001_03	IGNORED
C1	app	test_app_01	PASSED
C2	app	test_app_02	PASSED
C3	app_start	test_app_start_01	FAILED
C4	board_name	test_board_name_01	PASSED
C5	dht22_service	test_dht22_service_01	PASSED
C6	dht22_service	test_dht22_service_02	PASSED
C7	dht22_service	test_dht22_service_03	PASSED
C8	dht22_service	test_dht22_service_04	PASSED
C9	dht22_service	test_dht22_service_05	PASSED
C10	dht22_service	test_dht22_service_06	PASSED
C11	dht22_service	test_dht22_service_07	PASSED
C12	dht22_service_sample	test_dht22_service_sample_01	PASSED
C13	dht22_service_start	test_dht22_service_start_01	PASSED
C14	dht22_service_start	test_dht22_service_start_02	PASSED
C15	dht22_service_start	test_dht22_service_start_03	PASSED
C16	dht22_service_start	test_dht22_service_start_04	PASSED
C17	helpers	test_helpers_01	PASSED
C18	helpers	test_helpers_02	PASSED
C19	helpers	test_helpers_03	PASSED
C20	helpers	test_helpers_04	FAILED
C21	helpers	test_helpers_05	PASSED
C22	helpers	test_helpers_06	PASSED
C23	helpers	test_helpers_07	PASSED
C24	helpers	test_helpers_08	FAILED
C25	helpers	test_helpers_09	PASSED
C26	helpers	test_helpers_10	FAILED
C27	helpers	test_helpers_11	PASSED
C28	helpers	test_helpers_12	PASSED
C29	helpers	test_helpers_13	PASSED
C30	helpers	test_helpers_14	PASSED
C31	helpers	test_helpers_15	FAILED
C32	mpu6050	test_mpu6050_01	PASSED
C33	mpu6050	test_mpu6050_02	PASSED
C34	mpu6050	test_mpu6050_03	PASSED
C35	mpu6050	test_mpu6050_04	PASSED
C36	mpu6050	test_mpu6050_05	PASSED
C37	mpu6050	test_mpu6050_06	PASSED

Case	Identifier	Test function	Result
C38	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_01	PASSED
C39	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_02	PASSED
C40	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_03	PASSED
C41	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_04	PASSED
C42	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_05	PASSED
C43	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_06	PASSED
C44	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_07	PASSED
C45	mpu6050_configure_interrupt_pin	test_mpu6050_configure_interrupt_pin_08	PASSED
C46	mpu6050_deinit	test_mpu6050_deinit_01	PASSED
C47	mpu6050_deinit	test_mpu6050_deinit_02	PASSED
C48	mpu6050_get_who_am_i	test_mpu6050_get_who_am_i_01	PASSED
C49	mpu6050_handle_size	test_mpu6050_handle_size_01	PASSED
C50	mpu6050_init	test_mpu6050_init_01	PASSED
C51	mpu6050_init	test_mpu6050_init_02	PASSED
C52	mpu6050_init	test_mpu6050_init_03	PASSED
C53	mpu6050_init	test_mpu6050_init_04	FAILED
C54	mpu6050_init	test_mpu6050_init_05	PASSED
C55	mpu6050_init	test_mpu6050_init_06	PASSED
C56	mpu6050_init	test_mpu6050_init_07	PASSED
C57	mpu6050_init	test_mpu6050_init_08	PASSED
C58	mpu6050_init	test_mpu6050_init_09	PASSED
C59	mpu6050_init	test_mpu6050_init_10	PASSED
C60	mpu6050_init	test_mpu6050_init_11	PASSED
C61	mpu6050_init	test_mpu6050_init_12	PASSED
C62	mpu6050_init	test_mpu6050_init_13	PASSED
C63	mpu6050_init	test_mpu6050_init_14	PASSED
C64	mpu6050_init	test_mpu6050_init_15	PASSED
C65	mpu6050_init	test_mpu6050_init_16	PASSED
C66	mpu6050_read_fifo_count	test_mpu6050_read_fifo_count_01	PASSED
C67	mpu6050_read_register	test_mpu6050_read_register_01	PASSED
C68	mpu6050_service	test_mpu6050_service_01	PASSED
C69	mpu6050_service	test_mpu6050_service_02	PASSED
C70	mpu6050_service	test_mpu6050_service_03	PASSED
C71	mpu6050_service	test_mpu6050_service_04	PASSED

Case	Identifier	Test function	Result
C72	mpu6050_service	test_mpu6050_service_05	PASSED
C73	mpu6050_service	test_mpu6050_service_06	PASSED
C74	mpu6050_service	test_mpu6050_service_07	PASSED
C75	mpu6050_service	test_mpu6050_service_08	PASSED
C76	mpu6050_service	test_mpu6050_service_09	PASSED
C77	mpu6050_service	test_mpu6050_service_10	PASSED
C78	mpu6050_service	test_mpu6050_service_11	PASSED
C79	mpu6050_service	test_mpu6050_service_12	PASSED
C80	mpu6050_service	test_mpu6050_service_13	PASSED
C81	mpu6050_service	test_mpu6050_service_14	PASSED
C82	mpu6050_service	test_mpu6050_service_15	PASSED
C83	mpu6050_service_start	test_mpu6050_service_start_01	PASSED
C84	mpu6050_service_start	test_mpu6050_service_start_02	PASSED
C85	mpu6050_service_start	test_mpu6050_service_start_03	FAILED
C86	mpu6050_service_start	test_mpu6050_service_start_04	PASSED
C87	mpu6050_service_start	test_mpu6050_service_start_05	FAILED
C88	mpu6050_service_state	test_mpu6050_service_state_01	PASSED
C89	mpu6050_set_dlpf	test_mpu6050_set_dlpf_01	FAILED
C90	mpu6050_set_dlpf	test_mpu6050_set_dlpf_02	FAILED
C91	mpu6050_wake	test_mpu6050_wake_01	FAILED
C92	mpu6050_wake	test_mpu6050_wake_02	FAILED
C93	mpu6050_wake	test_mpu6050_wake_03	FAILED